

Table 3. Ablation Study of VLM Scale for Alignment Monitoring on R2R-CE (Val Unseen)

Monitor VLM	Params	Latency/Step (ms) ↓	VRAM (GB) ↓	NE ↓	SR (%) ↑	SPL (%) ↑
BLIP-2 (Ours)	~3B	~85	~8.5	6.08	26.4	17.3
InstructBLIP	~7B	~155	~14.2	6.02	26.8	17.5
LLaVA-1.5	~7B	~170	~15.0	6.00	26.7	17.7
Qwen-VL-Chat	~9.6B	~210	~18.5	5.95	27.1	17.6

Table 4. Fine-grained Sub-Instruction Evaluation: Comparison of Sub-Instruction Completion Rate (SICR) and Sub-Instruction Navigation Error (SINE) across different instruction complexities.

Dataset	Instruction Complexity	Method	Val Seen		Val Unseen	
			SICR (%) ↑	SINE (m) ↓	SICR (%) ↑	SINE (m) ↓
R2R-CE	Short (≤ 3 sub-instrs)	Baseline (Feed-forward)	78.4	2.15	71.2	2.89
		RefNav (Ours)	81.2	1.85	76.5	2.10
	Medium (4-5 sub-instrs)	Baseline (Feed-forward)	65.3	3.42	52.1	4.55
		RefNav (Ours)	77.8	2.10	68.4	3.05
	Long (≥ 6 sub-instrs)	Baseline (Feed-forward)	42.1	5.80	28.5	7.12
		RefNav (Ours)	69.4	3.25	59.2	4.60
RxR-CE	Short (≤ 3 sub-instrs)	Baseline (Feed-forward)	72.5	2.45	64.8	3.15
		RefNav (Ours)	75.6	2.12	69.2	2.65
	Medium (4-5 sub-instrs)	Baseline (Feed-forward)	58.6	4.10	44.3	5.25
		RefNav (Ours)	70.4	2.80	61.5	3.60
	Long (≥ 6 sub-instrs)	Baseline (Feed-forward)	35.8	6.90	22.4	8.45
		RefNav (Ours)	61.2	3.95	51.8	5.10